ROVER ELECTRICAL SYSTEM

BATTERY DIVIDER

VBAT
R1
TBD
VBAT_SENSE
R2
TBD
GND

BT1
20V / 4Ah BATTERY
GND

K1
MAIN DISCONNECT

VBAT

MOTOR BRANCH
SERVO BRANCH
ELECTRONICS
LiDAR / SENSORS

F1 TBD
F2 TBD
VBAT_SERVO_IN
F3 TBD
VBAT_LOGIC_IN
F4 TBD
VBAT_SENSOR_IN

U6
12V REGULATOR
VIN+
VOUT+
GND
+12V

U3
SERVO REGULATOR
V_SERVO_TBD

U4
5V REGULATOR
+5V

U5
SENSOR REGULATOR
V_SENSOR_TBD

MOTOR_L_PWM
MOTOR_L_DIR
MOTOR_R_PWM
MOTOR_R_DIR

U1
CYTRON LEFT DUAL
VBAT
GND
PWM_x2
DIR_x2
CH1_A
CH1_B
CH2_A
CH2_B

U2
CYTRON RIGHT DUAL

M1_A
M1_B
M2_A
M2_B
M3_A
M3_B
M4_A
M4_B

M1
LEFT FRONT
M2
LEFT REAR
M3
RIGHT FRONT
M4
RIGHT REAR

RASPBERRY PI 4
PWM_INPUT
A1

DRIVE LINK
SERVO LINK
LIDAR DATA LINK

PICO W – DRIVE
L_PWM
L_DIR
R_PWM
R_DIR
VSYS_(39)
A2
GND_(3)
ADC_TBD
VBAT_SENSE

PICO W – SERVOS
SERVO_1
SERVO_2
SERVO_3
SERVO_4
A3

SERVO_1_PWM
SERVO_2_PWM
SERVO_3_PWM
SERVO_4_PWM

S1 SERVO 1
S2 SERVO 2
S3 SERVO 3
S4 SERVO 4

LiDAR / SENSORS
V_SENSOR
A4